

Applied Industrial Organization

Classwork 3

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The database `blundell_bond_2000_production_function.dta` contains information about the production function of a sample of firms. In particular:

- `id` represents the id of the firm.
- `year`
- `sales`
- `labor` represents the number of employees
- `capital`

1. Create a Word file named `yourname_yoursurname_cw3.docx`
2. Create a new do-file and save it as `yourname_yoursurname_cw3.do`
3. Import the database in Stata.
4. **Descriptive questions.** Provide answers in the Word file:
 - (a) How many firms does the database contain?
 - (b) How many years? In which range (first and last year)?
5. Plot the distribution of the variables `sales`, `labor` and `capital`.
6. Find an appropriate graph to show the relation between `sales` and `labor`.
7. Now, you assume that the production function can be approximated using a Cobb-Douglas with labor (L) and capital (K) as inputs:

$$Y = AL^{\alpha_L} K^{\alpha_K}$$

estimate the appropriate regression assuming that the *Total Factor Productivity* (A) is independent from the two inputs L and K .

8. Then, we want to relax the independence assumption, so we review the following estimation methods:

- (a) Input prices as IVs
- (b) Firm-Fixed Effects estimator
- (c) Firm+Year-Fixed Effects estimator
- (d) Fixed Effects + Cochrane-Orcutt estimator

Which of the following estimators can be implemented with the available information? Why?

9. Estimate all the above models that can be estimated using the available information. Which one seems more reliable?
10. Starting from the model that you selected in the previous point, briefly comment on the **Returns of Scale** of the analysed industry.
11. Now, you want to see if alternative production functions may better fit. For example, you want to try to obtain the parameters of the following production function:

$$Y = \exp \left(\alpha \frac{K}{L} \right)$$

obtain the estimation equation and estimate it. Does the model seem reasonable? Why?